
Final Project Report MATH 563

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Abstract

give some background

1 Introduction

1.1 Notation

1.2 Preliminaries

We first note an important relation between the proximal and subdifferential operators.

Proposition 1. Given a function $f \in \Gamma$, we have

$$\text{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \quad (1)$$

where $(I + \lambda \partial f)^{-1}$, for $\lambda > 0$, is called the **resolvent** of the operator ∂f . In particular, the proximal operator is the resolvent of the subdifferential operator [3].

Definition 1 (Maximal Monotone operator). A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If A is monotone (resp. maximal monotone) then so are A^{-1} and λA if $\lambda > 0$.

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

Proposition 2. If F is maximal monotone, then $(I + \lambda F)^{-1}(\hat{x})$ exists and is unique for all $\hat{x} \in \mathbb{R}^n$. The value $x = (I + \lambda F)^{-1}(\hat{x})$ of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x) \quad (2)$$

Proposition 3 (Proximal operator splitting). If f is separable across two variables, so $f(x, y) = \phi(x) + \psi(y)$, then

$$\text{prox}_f(x, y) = (\text{prox}_\phi(x), \text{prox}_\psi(y)) \quad (3)$$

Thus, evaluating the proximal operator of a separable function reduces to evaluating the proximal operators for each of the separable parts, which can be done independently [3].

2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$\begin{aligned} & \min_{x, y} f(x) + g(y) \\ & \text{subject to } Ax = y, \end{aligned} \quad (4)$$

where $f, g \in \Gamma$ have inexpensive prox-operators. We are given a vectorized, blurred image b , along with the kernel K used to blur it using the model

$$Kx + \eta = b, \quad (5)$$

where x is the original image, and η the added noise. Our goal is to recover the original image x . We formulate the problem in two ways, namely, one which uses the L^1 norm, and the other the L^2 norm. First,

$$\min_x \|Kx - b\|_1 + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \quad (6)$$

and second,

$$\min_x \|Kx - b\|_2^2 + \delta_S(x) + \gamma \|Dx\|_{\text{iso}}, \quad (7)$$

where $S = \{x \mid 0 \leq x \leq 1\}$, so that δ_S aims to restrict the range of allowed pixels. The iso norm is defined as

$$\|(x, y)\|_{\text{iso}} = \sum_{k=1}^n (x_k^2 + y_k^2)^{\frac{1}{2}}$$

2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (6), let

$$\begin{aligned} f(x) &= \delta_S(x) \\ g(x) &= \|Kx - b\|_1 + \gamma \|Dx\|_{\text{iso}} \end{aligned} \quad (8)$$

and consider the minimization problem

$$\min_{x,y} f(x) + g(y) + \delta_0 \left([A \quad -I] \begin{bmatrix} x \\ y \end{bmatrix} \right) \quad (9)$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0 \left([A \quad -I] \begin{bmatrix} x \\ y \end{bmatrix} \right) =: F(x, y) \quad (10)$$

As stated in the literature [2], F is the subdifferential of closed, convex functions, hence is a maximal monotone operator. Thus, by (2), the zeros of F are the fixed points of the resolvent of F . Therefore, if we can find the fixed points of F , we'll have found the optimal points. Unfortunately, this particular problem is too computationally expensive, so the strategy is to split F into two maximal monotone operators A and B which have inexpensive proximal operators. Finding the fixed points of $A + B$ will be equivalent to finding that of F . Define A and B as,

$$\begin{aligned} A(x, y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky - b\|_1 + \gamma \|Dy\|_{\text{iso}}) \\ B(x, y) &= \partial \delta_0 \left([A \quad -I] \begin{bmatrix} x \\ y \end{bmatrix} \right) \end{aligned} \quad (11)$$

We now focus on finding the resolvents of A and B ,

$$\begin{aligned} J_A &= (I + \lambda A)^{-1} = \text{prox}_{\lambda(f+g)}(x, y) \quad [\text{by (1)}] \\ &= (\text{prox}_{\lambda f}(x), \text{prox}_{\lambda g}(y)) \quad [\text{by Prop. 3}] \end{aligned} \quad (12)$$

where

$$\text{prox}_{\lambda f}(x) = \text{prox}_{\lambda \delta_S}(x) = \arg \min_x \frac{1}{2} \|u - x\|_2^2 + \delta_S(x) = \arg \min_x \frac{1}{2} \|u - x\|_2^2, \quad (13)$$

which is the euclidean projection operator onto S . Now, we observe that $g(y)$ described in (8) can be written in a different form by letting

$$y_1 = Ky \quad \text{and} \quad (y_2, y_3) = Dy$$

to get

$$g(y_1, y_2, y_3) = \|y_1 - b\|_1 + \gamma \|(y_2, y_3)\|_{\text{iso}} \quad (14)$$

Then, by Proposition 3, we notice that the proximal operator of g may be obtained by splitting its components into

$$\varphi(y_1) = \|y_1 - b\|_1 \quad \text{and} \quad \psi(y_2, y_3) = \gamma \|(y_2, y_3)\|_{\text{iso}}$$

which gives

$$\mathbf{prox}_{\lambda g}(y_1, y_2, y_3) = (\mathbf{prox}_{\lambda \varphi}(y_1), \mathbf{prox}_{\lambda \psi}(y_2, y_3)) \quad (15)$$

So, we find

$$\mathbf{prox}_{\lambda \varphi}(y_1) = \mathbf{prox}_{\lambda \|\cdot - b\|_1}(y_1)$$

Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

Alexandre St-Aubin (Student ID: 261115785) We ball.

Jonathan Campana (Student ID:) We ball.

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References

- [1] Jim Douglas and H. H. Rachford. On the numerical solution of heat conduction problems in two and three space variables. *Transactions of the American Mathematical Society*, 82(2):421, Jul 1956. doi: <https://doi.org/10.2307/1993056>. URL <https://typeset.io/pdf/on-the-numerical-solution-of-heat-conduction-problems-in-two-41bdiqc5i8.pdf>.
- [2] Daniel O'Connor and Lieven Vandenbergh. Primal-dual decomposition by operator splitting and applications to image deblurring. *SIAM Journal on Imaging Sciences*, 7(3):1724–1754, 2014. doi: 10.1137/13094671X. URL <https://doi.org/10.1137/13094671X>.
- [3] Neal Parikh. Proximal algorithms. *Foundations and Trends in Optimization*, 1(3):127–239, 2014. doi: <https://doi.org/10.1561/24000000003>.

A Appendix

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.